

Numerical evidence for KPZ universality

Discrete growth models and a spectral IMEX solver

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Abstract

We study one-dimensional kinetic roughening using three discrete growth models – random deposition with surface relaxation (RDSR), ballistic deposition (BD), and restricted solid-on-solid (RSOS) – and a continuous Kardar–Parisi–Zhang (KPZ) equation integrated with a spectral implicit-explicit (IMEX) scheme. The interface width is measured over many independent realizations, and the roughness, growth, and dynamic exponents are estimated from temporal growth, finite-size saturation, and Family–Vicsek data collapse. RDSR reproduces the Edwards–Wilkinson (EW) exponents, while RSOS clearly reproduces the KPZ class. BD has the correct growth exponent but its direct finite-size power-law fit is strongly biased, highlighting the importance of corrections to scaling. The continuous solver gives a roughness exponent close to KPZ and a noisier estimate of the dynamic exponent. The notebook also establishes a reproducible workflow for noise generation, parameter control, uncertainty estimates, and bootstrap resampling.

1 Introduction and theoretical framework

Growth models describe the emergence of irregular interfaces in processes such as deposition, erosion, fires, and colony expansion. Their microscopic rules can be very different while their large-scale fluctuations are governed by a small number of universality classes. The present work compares discrete simulations with a continuous numerical solution of KPZ dynamics, with emphasis on exponent estimates and scaling collapse.

The EW class is defined in the notebook by

$$\begin{aligned} \mathcal{C}_{EW} &:= \{h : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R} \mid h \Rightarrow h_{EW}\}, \\ \partial_t h_{EW} &= \nu \nabla^2 h_{EW} + \eta(D), \quad \nu > 0, \ D > 0, \\ \{\alpha, \beta, z\}(d=1) &= \left\{ \frac{1}{2}, \frac{1}{4}, 2 \right\} \in \mathcal{C}_{EW}. \end{aligned} \tag{1}$$

The KPZ class is defined analogously by

$$\begin{aligned} \mathcal{C}_{KPZ} &:= \{h : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R} \mid h \Rightarrow h_{KPZ}\}, \\ \partial_t h_{KPZ} &= \nu \nabla^2 h_{KPZ} + \frac{\lambda}{2} (\nabla h_{KPZ})^2 + \eta(D), \\ \nu > 0, \ D > 0, \ \lambda &\neq 0, \\ \{\alpha, \beta, z\}(d=1) &= \left\{ \frac{1}{2}, \frac{1}{3}, \frac{3}{2} \right\} \in \mathcal{C}_{KPZ}. \end{aligned} \tag{2}$$

For an interface of lateral size L , the measured roughness is

$$W(L, t) = \sqrt{\langle (h - \langle h \rangle)^2 \rangle}. \tag{3}$$

The Family–Vicsek form used throughout the project is

$$W(L, t) = L^\alpha f\left(\frac{t}{L^z}\right), \quad f(u) \sim \begin{cases} u^\beta, & u \ll 1, \\ \text{const.}, & u \gg 1, \end{cases} \quad \beta = \frac{\alpha}{z}. \tag{4}$$

Thus, β is obtained from the growth regime, α from the saturated width $W_{\text{sat}}(L)$, and z from α/β . Exponents alone are necessary but not sufficient evidence for universality: the rescaled curves W/L^α versus t/L^z must also collapse. The notebook records two complementary theoretical viewpoints. First, coarse-graining over a scale l is represented by

$$\mathcal{R}_l[h](x) = \frac{1}{l^d} \int_{|y-x|<l} h(y) \, dy, \tag{5}$$

which connects KPZ fluctuations to an EW-like large-scale description in the appropriate regime. Second, for Gaussian noise of variance $2\mathcal{D}$, the KPZ dynamics can be written using the Martin–Siggia–Rose–Janssen–De Dominicis action

$$S[h, \tilde{h}] = \int dt \, d^d x \left[\tilde{h} \left(\partial_t h - \nu \nabla^2 h - \frac{\lambda}{2} (\nabla h)^2 \right) - \mathcal{D} \tilde{h}^2 \right]. \tag{6}$$



Figure 1: Examples used in the project to motivate interface-growth models: burning vegetation, bacterial growth, coastal erosion, deposition, porous materials, and water-wave patterns.

The response field enforces the stochastic dynamics, while the quadratic term contains the noise correlations. These equations motivate the scaling analysis; the numerical work below tests the scaling directly.

2 Discrete models

2.1 Microscopic rules

All discrete simulations use a height field on a one-dimensional periodic lattice. The implementation uses zero-based indices, while the mathematical statements below retain the notebook notation $\{1, \dots, L\}$. At each step a site is selected uniformly, and the height is updated according to the local rule.

For RDSR, deposition occurs at the lowest site among the selected site and its two neighbors:

$$\begin{cases} h : \{1, \dots, L\} \times \mathbb{N} \rightarrow \mathbb{Z}, \\ \mathbb{P}(h \rightarrow h + e_j) = \frac{\text{card}\{i : j = \arg \min_{k \in \{i-1, i, i+1\}} h_k\}}{L}, \\ \text{left}_i = \begin{cases} i-1, & i > 0, \\ L_x - 1, & \text{otherwise,} \end{cases} \quad \text{right}_i = \begin{cases} i+1, & i < L_x - 1, \\ 0, & \text{otherwise.} \end{cases} \end{cases} \quad (7)$$

For BD, a selected column sticks immediately above the local maximum:

$$\begin{cases} h : \{1, \dots, L\} \times \mathbb{N} \rightarrow \mathbb{Z}, \\ \mathbb{P}(h \rightarrow h + e_j) = \frac{\text{card}\{i : j = \arg \max_{k \in \{i-1, i, i+1\}} h_k\}}{L}, \\ \text{left}_i = \begin{cases} i-1, & i > 0, \\ L_x - 1, & \text{otherwise,} \end{cases} \quad \text{right}_i = \begin{cases} i+1, & i < L_x - 1, \\ 0, & \text{otherwise.} \end{cases} \end{cases} \quad (8)$$

For RSOS, a deposition is accepted only if the height difference to both neighbors remains at most one:

$$\begin{cases} h : \{1, \dots, L\} \times \mathbb{N} \rightarrow \mathbb{Z}, \\ \mathbb{P}(h \rightarrow h + e_j) = \frac{\text{card}\{i : |h_i + 1 - h_k| \leq 1, \forall k \in \{i-1, i, i+1\}\}}{L}, \\ \text{left}_i = \begin{cases} i-1, & i > 0, \\ L_x - 1, & \text{otherwise,} \end{cases} \quad \text{right}_i = \begin{cases} i+1, & i < L_x - 1, \\ 0, & \text{otherwise.} \end{cases} \end{cases} \quad (9)$$

2.2 Sampling and exponent estimates

For each model, $W(t)$ is evaluated during a realization and averaged over independent runs. The growth exponent is the slope of a log-log regression, $\log W = \beta \log t + c$, restricted to a pre-saturation window. We used 40 independent

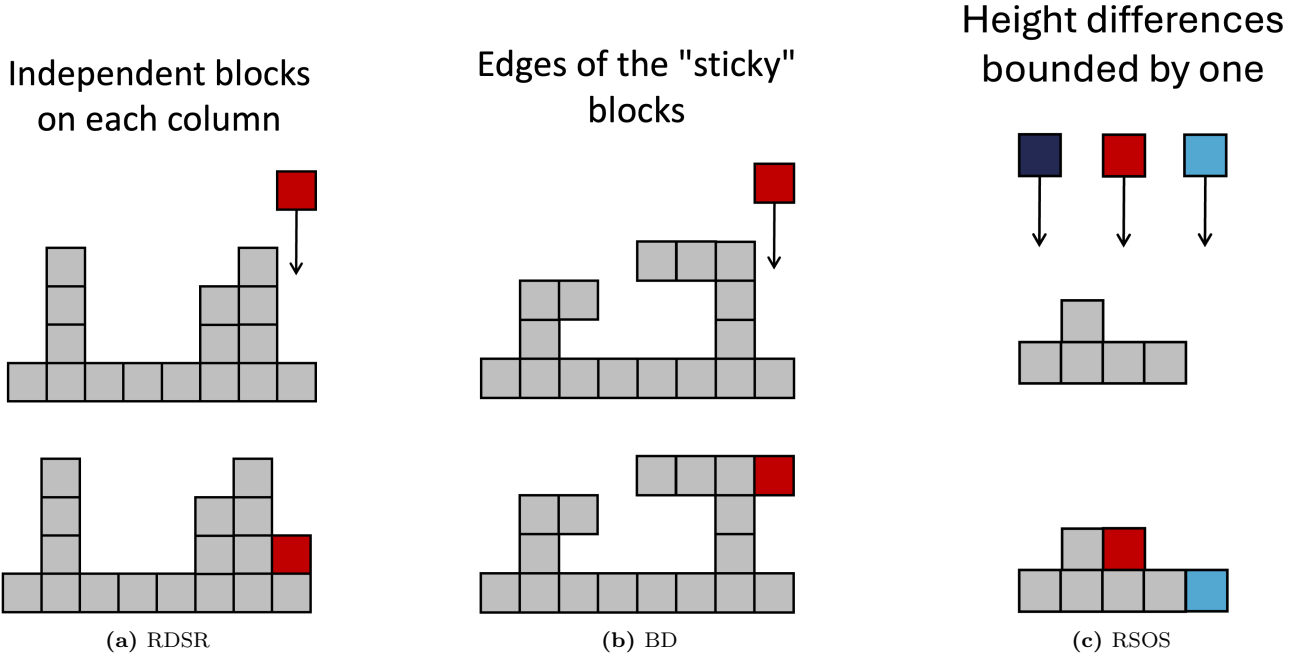


Figure 2: Microscopic deposition sketches reused from the project notebook. The highlighted site indicates the selected update according to each rule.

realizations with seeds $42, \dots, 81$: $L = 512$ for RDSR and BD, and $L = 4096$ for RSOS. The corresponding windows were $1 < t < 0.2t_{\max}$ for RDSR and IMEX, $1 < t < 0.9t_{\max}$ for BD, and $80000 < t < 0.9t_{\max}$ for RSOS.

The uncertainty of the mean growth exponent is the standard error of the mean,

$$\text{SEM} = \frac{\text{std}\{\beta\}}{\sqrt{n_{\text{runs}}}}. \quad (10)$$

For the finite-size analysis, 500 runs were averaged for each size. The saturation width was estimated from the final plateau portion of the averaged signal, and a weighted linear fit of $\log W_{\text{sat}}$ versus $\log L$ yielded α with the covariance of the fit. The discrete sizes were $L = 16, 32, 64, 128$ for RDSR and $L = 16, 32, 64, 128, 256$ for BD and RSOS. The number of steps was chosen so that the simulation extended well beyond saturation, following the notebook criterion $n_{\text{steps}} \sim 5L^{z+1}$.

BD is the most delicate case. Its direct fit gives $\alpha = 0.359 \pm 0.006$, far below the KPZ value. The project therefore introduces the finite-size correction

$$W_{\text{sat,model}}(L) = AL^\alpha [1 + BL^{-\omega}]. \quad (11)$$

The associated bootstrap procedure resamples rows of the matrix

$$\mathbf{W}_{\text{sat}} \in \mathbb{R}^{N_{\text{runs}} \times N_L}, \quad \langle W_{\text{sat}}(L_j) \rangle = \frac{1}{N_{\text{runs}}} \sum_{i=1}^{N_{\text{runs}}} W_{\text{sat}}^{(i)}(L_j), \quad (12)$$

with replacement. Each resampled mean is fitted to Eq. (11), producing $\{\alpha_b\}_{b=1}^{N_{\text{boot}}}$ and

$$\sigma_\alpha = \left[\frac{1}{N_{\text{boot}} - 1} \sum_{b=1}^{N_{\text{boot}}} (\alpha_b - \bar{\alpha})^2 \right]^{1/2}. \quad (13)$$

This is the statistically preferred route for BD and for future measurements. The final notebook summary retains the direct power-law estimate, so the finite-size bias remains explicit.

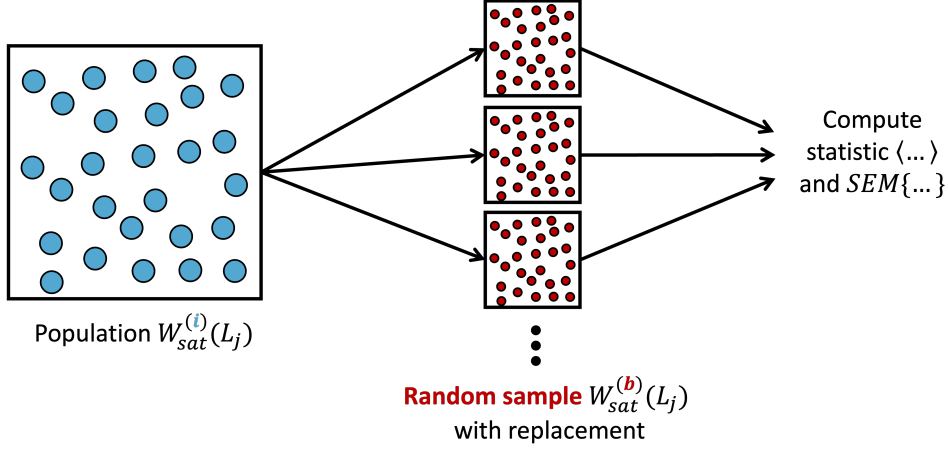


Figure 3: Bootstrap resampling scheme used to motivate the uncertainty analysis for finite-size fits.

3 Continuous KPZ equation and spectral IMEX method

The continuous model is solved on $[0, L_x]$ with N_x Fourier modes and periodic boundary conditions. The notebook formulation is

$$\begin{cases} h : [0, L_x] \times \mathbb{N} \rightarrow \mathbb{R}, \\ \partial_t h(x, t) = \nu \partial_x^2 h(x, t) + \frac{\lambda}{2} (\partial_x h(x, t))^2 + \eta(x, t), \\ \eta(x, t) \sim \mathcal{N}\left(0, \frac{2D}{\Delta x \Delta t}\right), \\ \hat{h}_k^{n+1} = \frac{\hat{h}_k^n + \Delta t (\widehat{NL}_k^n + \hat{\eta}_k^n)}{1 + \Delta t \nu k^2}, \\ h(0, t) = h(L_x, t). \end{cases} \quad (14)$$

The implicit Laplacian makes the update unconditionally stable for the linear diffusion part, while the nonlinear term and noise are explicit. The Fourier wave numbers are generated as

$$k = 2\pi \text{fttfreq}(N_x, d = \Delta x), \quad \Delta x = \frac{L_x}{N_x - 1}. \quad (15)$$

The nonlinear convolution is dealiased with the standard 2/3 rule. The discrete noise is generated from independent standard Gaussians as

$$\eta_i^n = \sqrt{\frac{2D}{(\Delta x)^d \Delta t}} R_i^n, \quad R_i^n \sim \mathcal{N}(0, 1), \quad (16)$$

which gives $\langle \eta_j^n \eta_{j'}^{n'} \rangle = 2D(\Delta x \Delta t)^{-1} \delta_{j,j'} \delta_{n,n'}$ in one dimension. Optional regularization replaces $(\nabla h)^2$ by

$$f((\nabla h)^2) = \frac{1 - \exp[-c(\nabla h)^2]}{c}, \quad c > 0, \quad (17)$$

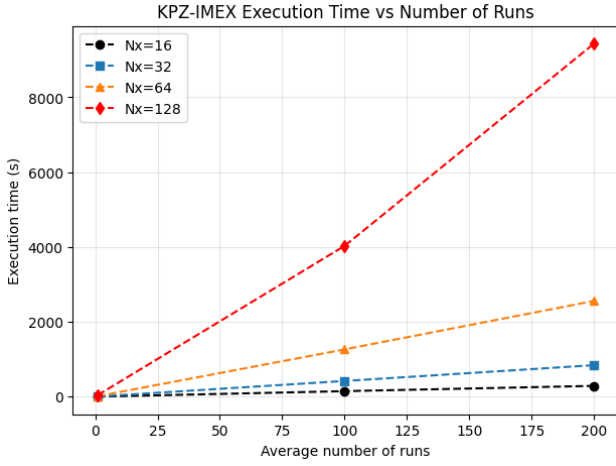
to prevent large gradients from causing numerical blow-up. In the reported simulations dealiasing was enabled and regularization was disabled.

For the finite-size study, $\nu = 1$, $\lambda = 1$, $D = 0.2$, $\Delta t = 0.01$, $L_x = N_x - 1$, and $N_x = 16, 32, 64, 128$. The step counts were 32000, 90510, 256000, and 724077, with output every 30 steps and 200 realizations per size. The growth exponent used a separate $N_x = 512$, $L_x = 511$, $D = 1$ configuration with 40 realizations, 5000 steps, and output every 250 steps. The measured vectors of times are identical across runs at fixed size because the time grid is deterministic.

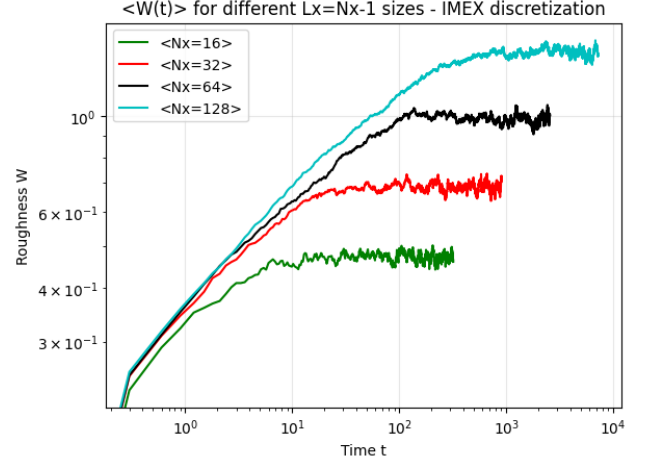
4 Results

4.1 Measured exponents

Table 1 gathers the final values reported by the notebook. The uncertainties on β are SEM values, those on α are from the weighted fit covariance, and the z uncertainties are propagated from $z = \alpha/\beta$. The notebook also derives a covariance-aware propagation formula; the final numerical output uses the implemented expression.



(a) Runtime scaling



(b) Growth and saturation

Figure 4: Continuous KPZ diagnostics from the spectral IMEX implementation. The rapid growth in runtime limited the finite-size study to $N_x \leq 128$.

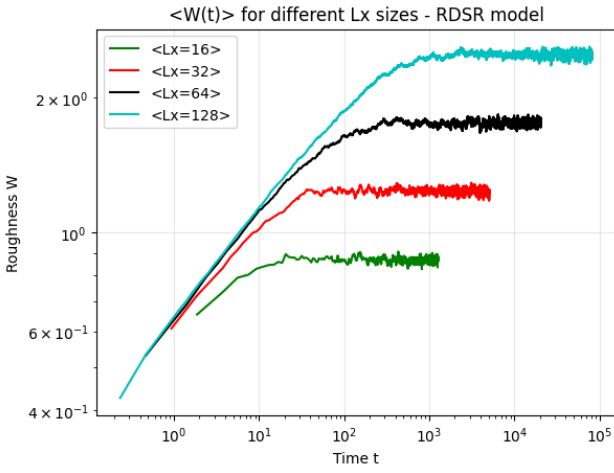
Table 1: Measured one-dimensional exponents. The expected values are $\{\alpha, \beta, z\}_{EW} = \{0.5, 0.25, 2\}$ and $\{\alpha, \beta, z\}_{KPZ} = \{0.5, 1/3, 1.5\}$.

Model	Class	α	β	$z = \alpha/\beta$	Main diagnostic
RDSR	EW	0.509 ± 0.002	0.254 ± 0.004	2.016 ± 0.182	exponents + collapse
BD	KPZ	0.359 ± 0.006	0.335 ± 0.007	1.071 ± 0.151	β only reliable
RSOS	KPZ	0.509 ± 0.001	0.329 ± 0.005	1.547 ± 0.151	exponents + collapse
KPZ-IMEX	KPZ	0.511 ± 0.003	0.293 ± 0.011	1.745 ± 0.408	α + collapse

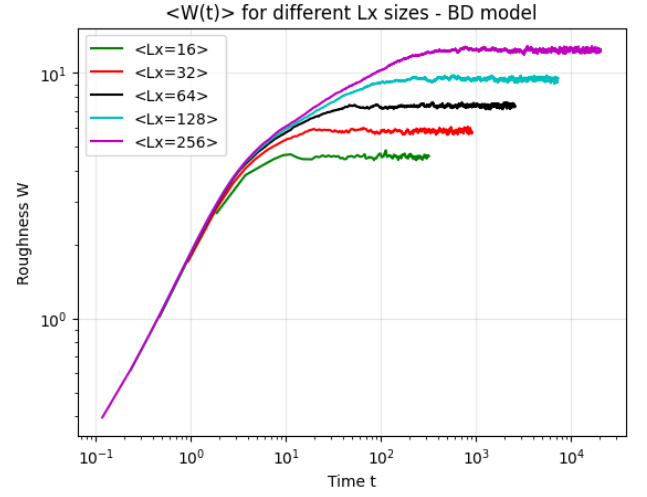
RDSR agrees with the EW class at the level of 1.47% for β , 1.71% for α , and 0.78% for z . Its averaged growth curves saturate at size-dependent widths, and the rescaling in Eq. (4) collapses the data over the growth-to-saturation crossover (Fig. 7, top left). This is the cleanest numerical EW result.

BD gives an accurate growth exponent, $\beta = 0.335 \pm 0.007$, but the raw finite-size fit gives a 28.23% deviation in α . The failure is visible as a finite-size bias rather than as a failure of the local KPZ rule: the growth curves have the expected KPZ-like power-law sector, while the direct $W_{\text{sat}} \sim L^\alpha$ fit is too sensitive to the limited size range and corrections to scaling. Equation (11) and the bootstrap construction in Fig. 3 are the appropriate methodological response. The notebook’s final table deliberately reports the uncorrected estimate, with z consequently also biased. RSOS reproduces all three KPZ exponents closely: the deviations from $\{0.5, 1/3, 1.5\}$ are 1.72%, 1.35%, and 3.11% for α , β , and z , respectively. The growth curves, finite-size fit, and collapse are mutually consistent.

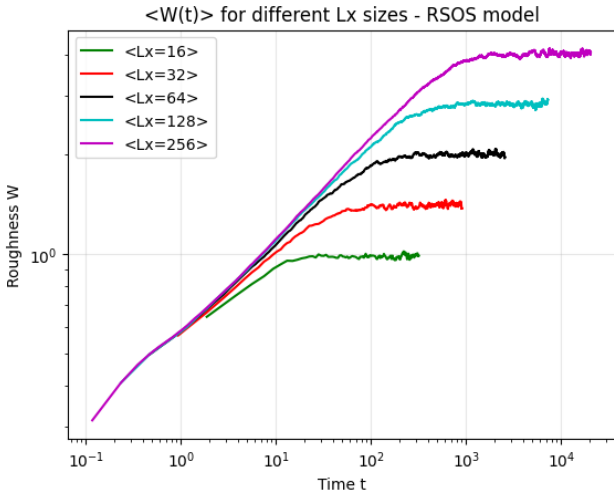
For the continuous solver, $\alpha = 0.511 \pm 0.003$ is close to the KPZ prediction, whereas $\beta = 0.293 \pm 0.011$ is lower by 12.22%. The resulting $z = 1.745 \pm 0.408$ has a large uncertainty. The collapse is good from the intermediate growth regime into saturation, with the expected deviations at very small t/L^z where discretization transients dominate. The IMEX result is therefore supportive, but less statistically sharp than RSOS.



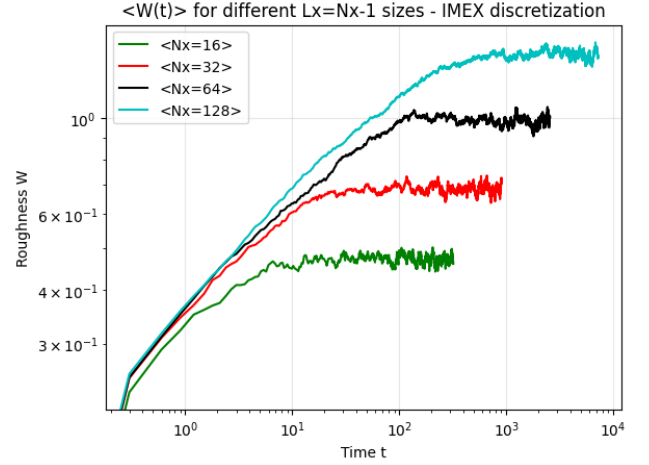
(a) RDSR



(b) BD



(c) RSOS

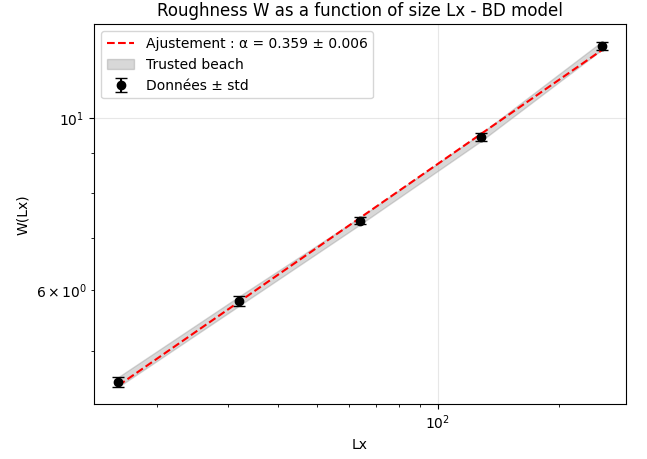


(d) KPZ-IMEX

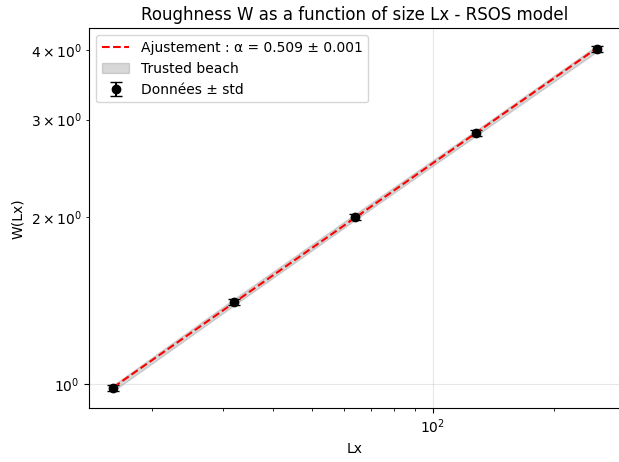
Figure 5: Averaged roughness $\langle W(t) \rangle$ for increasing system sizes. The curves show the growth regime followed by finite-size saturation.



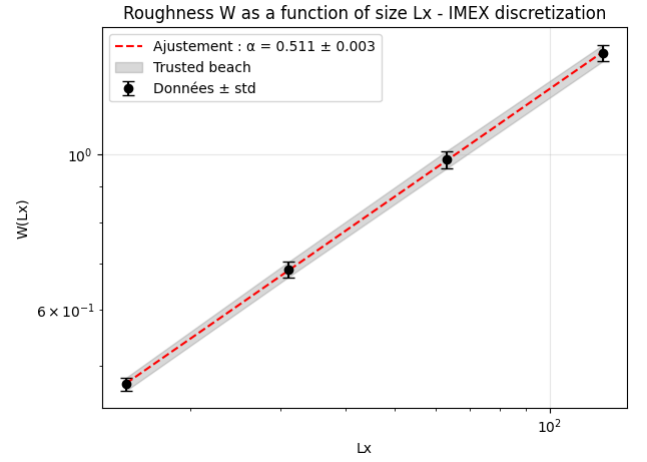
(a) RDSR: $\alpha = 0.509 \pm 0.002$



(b) BD: $\alpha = 0.359 \pm 0.006$

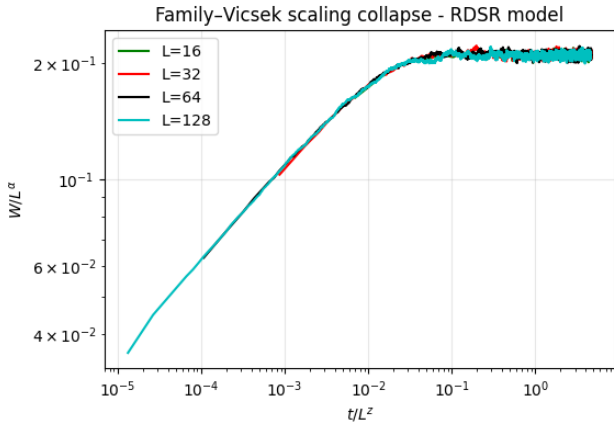


(c) RSOS: $\alpha = 0.509 \pm 0.001$

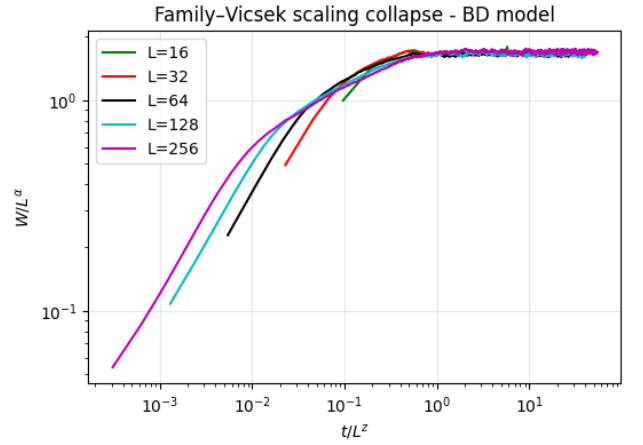


(d) KPZ-IMEX: $\alpha = 0.511 \pm 0.003$

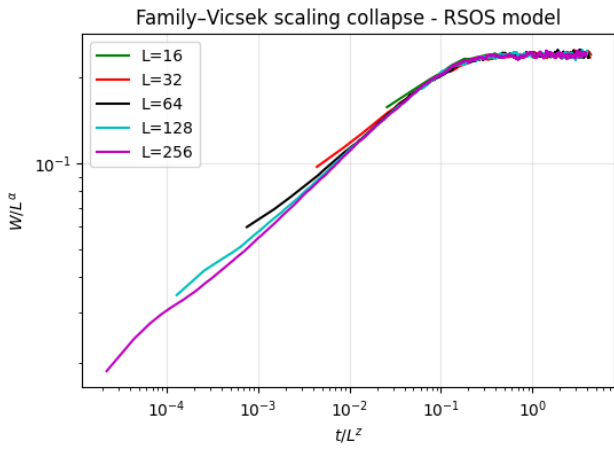
Figure 6: Finite-size saturation widths and weighted power-law fits. The BD panel illustrates the finite-size bias that motivates Eq. (11).



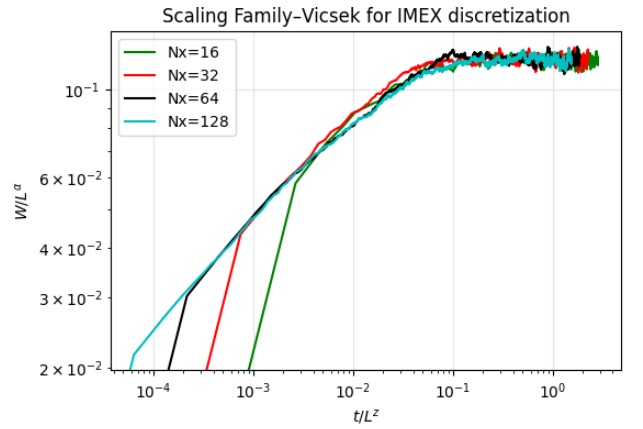
(a) RDSR, EW scaling



(b) BD, raw-fit scaling



(c) RSOS, KPZ scaling



(d) KPZ-IMEX scaling

Figure 7: Family-Vicsek data collapse: W/L^α against t/L^z . The intermediate and saturation regimes collapse well; the earliest points are pre-asymptotic.

5 Discussion and conclusion

The project establishes a coherent numerical comparison between discrete and continuous descriptions of interface growth. RDSR provides a controlled EW benchmark. RSOS gives a particularly clear KPZ realization, with all measured exponents and its scaling collapse in agreement with the one-dimensional KPZ predictions. BD confirms the KPZ growth exponent but also shows why exponent matching must be accompanied by finite-size analysis: a direct power law over a restricted range can give a severely biased roughness exponent. The proposed correction-to-scaling fit and bootstrap resampling preserve the run-by-run structure and are the natural next step for a robust BD estimate.

The discrete simulations benefited from Numba just-in-time compilation, which made hundreds of realizations and relatively large lattice sizes practical. The continuous solver cannot use the same acceleration path, and its FFT/IFFT cost grows rapidly: the recorded timings increase from 0.9 s for one $N_x = 16$ run to 40.7 s for one $N_x = 128$ run, and from 2 min 27 s to 67 min 9 s for 100 runs. This computational constraint explains the smaller ensemble and size range of the IMEX study.

The uncertainty treatment is adequate for a concise numerical study: SEM is used for independent growth-exponent estimates, fit covariances are retained for weighted finite-size regressions, and a bootstrap workflow is specified for nonlinear corrections. A quantitative global distance between collapsed curves would strengthen the universality claim beyond visual inspection. Other natural extensions are a fully converged bootstrap analysis, faster spectral backends, ETDRK4 or Cole–Hopf-based integrators, kinetic Ising-type KPZ models, and a carefully averaged two-dimensional study. The two-dimensional case was intentionally not pursued here because it would require substantially more computation and does not have the same analytic exponent benchmark.

References

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- [2] F. Family and T. Vicsek, “Scaling of the active zone in the Eden process on percolation networks and the ballistic deposition model,” *Journal of Physics A* **18**, L75–L81 (1985).
- [3] M. Kardar, G. Parisi, and Y.-C. Zhang, “Dynamic scaling of growing interfaces,” *Physical Review Letters* **56**, 889–892 (1986).